

HW3 Conversation History

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{"step_index":0,"source":"USER_EXPLICIT","type":"USER_INPUT","status":"DONE","created_at":"2026-05-13T09:02:36Z","content":"\n我要完成這四個作業\n其中3-1. \n結果要有\nloss curve\nreward curve\n測試結果\n3-2. \n結果要比較\nBasic DQN\nDouble DQN\nDueling DQN\n的loss curve\nreward curve\n測試結果\n3-3. \n選 PyTorch Lightning\n\n加入1. Gradient clipping\n2. Learning rate scheduler\n3. Target network update\n結果要有\nloss curve\nreward curve\n測試結果\n3-4. \n結果要有\nloss curve\nreward curve\n測試結果\n\n幫我完成上述\n\n\nThe current local time is: 2026-05-13T17:02:36+08:00.\n\nThe user's current state is as follows:\nActive Document: /home/william/Desktop/chdr/hw3/hw3_2.py (LANGUAGE_PYTHON)\nCursor is on line: 1\nOther open documents:\n- /home/william/Desktop/chdr/hw3/hw3_2.py (LANGUAGE_PYTHON)\n- /home/william/Desktop/chdr/hw3/hw3_1.py (LANGUAGE_PYTHON)\n\nNo browser pages are currently open.\n\n\n"}
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2. 3-2: 修改 hw3_2.py 以依次訓練 Basic DQN, Double DQN 和 Dueling DQN，並將它們的 Loss 和 Reward 曲線畫在同一張圖上進行比較，最後測試這三個模型。
3. 3-3: 將 hw3_3.py 的 Keras 程式碼改寫為 PyTorch Lightning，並依要求加入 Gradient clipping、Learning rate scheduler 與 Target network update。同樣也會輸出曲線和測試結果。
\n\n不過，關於 3-4，您只提到需要「loss curve、reward curve、測試結果」，但沒有說明 3-4 具體要解決什麼問題（例如：是要使用哪種模型架構？還是要在哪一種環境 mode 下測試？或是要實作什麼特定的 baseline？）\n\n"}
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{"step_index":30,"source":"MODEL","type":"PLANNER_RESPONSE","status":"DONE","created_at":"2026-05-13T09:06:48Z","content":"我已經看"}

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您提供的圖片，並將 HW3-4 的內容更新到側邊欄的 `implementation_plan.md` 實作計畫中了。
 \n\n**針對 Rainbow DQN 的分析：**
 Rainbow DQN 是一篇在強化學習領域非常經典的論文，它結合了六個對基本 DQN 的改良技術，讓模型在各種環境下都能達到最先進 (SOTA) 的表現：
 \n1. **Double DQN:** 解決 Q 值高估的問題。（我們在 3-2 會實作）
 \n2. **Dueling DQN:** 將網路拆分為狀態價值 (Value) 與優勢函數 (Advantage)，幫助模型更好地區分不同狀態的好壞。（我們在 3-2 會實作）
 \n3. **Prioritized Experience Replay (PER):** 不再是隨機從 Buffer 抽取經驗，而是根據 TD Error (預測誤差) 的大小來決定抽樣機率，誤差越大的經驗越有學習價值，抽到的機率越高。
 \n4. **Multi-step Learning (N-step):** 不只看下一步的 Reward，而是看未來 N 步的 Reward 總和，這能加快學習速度並平衡 Bias 與 Variance。
 \n"}

[illegible]

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import deque\nimport copy\nimport matplotlib.pyplot as plt\n<truncated 7849 bytes>","Description":"\Rewrite hw3_3.py to use PyTorch
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1\n","EndLine":"169","Instruction":"\Change DummyDataset(self.epochs) to DummyDataset(1) so PyTorch Lightning epochs match our
episodes.\n","ReplacementContent":"\def train_dataloader(self):\n    dataset = DummyDataset(1)\n    return
DataLoader(dataset, batch_size=1)\n","StartLine":"167","TargetContent":"\def train_dataloader(self):\n    dataset =
DummyDataset(self.epochs)\n    return DataLoader(dataset,
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[X]Rewritehw3_3.pyusing PyTorch Lightning.\n -[X]Add Gradient Clipping.\n -[X]Add Learning Rate Scheduler.\n -[X]Add Target
Network Update.\n -[X]P\n<truncated 312 bytes>","TargetFile":"\home/william/.gemini/antigravity/brain/7319aa41-c77b-4380-8b23-
7cf9268d040a/task.md\n"}]]}

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np\nimport random\nfrom collections import deque\nimport copy\nimport matplotlib.pyplot as plt\nfrom Gridworld import
Gridworld\n\n\naction_set = {0: 'u', 1: 'd', 2: 'l', 3: 'r'\n<truncated 9578 bytes>","Description":"\Create hw3_4.py with Simplified
Rainbow DQN\n","IsArtifact":"false","Overwrite":"false","TargetFile":"\home/william/Desktop/chdr1/hw3/hw3_4.py\n"}]]}

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[X]Createhw3_4.py.\n -[X]Implement simplified Rainbow DQN (Double + Dueling + PER + Multi-step).\n -[X]Train onrandommode and
track metrics.\n -[X]P\n<truncated 440 bytes>","TargetFile":"\home/william/.gemini/antigravity/brain/7319aa41-c77b-4380-8b23-
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of changes made to complete HW3\n"},"CodeContent":"\n# HW3 Completion Walkthrough\n\nI have successfully completed all four parts of
your Deep Reinforcement Learning homework. Below is a detailed walkthrough of the changes and achievements for each assignment.\n\n\n#

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HW3-1: Basic DQN in Static Mode\\n**Goal**\\n<truncated 3131 bytes>","Description":"Created walkthrough for
HW3\\n","IsArtifact":"true","Overwrite":"false","TargetFile":"\\n/home/william/.gemini/antigravity/brain/7319aa41-c77b-4380-8b23-
7cf9268d040a/walkthrough.md\\n"}]]}

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成了全部四個部分的作業要求，並且都在您的環境中執行測試完畢了！您可以在右側的walkthrough.md看到完整的執行結果摘要。\\n\\n這裡幫您總結一下四個檔案的修改與成果：
\\n1. **3-1 (hw3_1.py)**: 加入了 Reward 紀錄機制，會在訓練結束後儲存loss_curve_3_1.png與reward_curve_3_1.png，並執行 10 局無探索的測試，測試勝
率達 100%。\\n2. **3-2 (hw3_2.py)**: 將程式碼改寫為依次訓練 Basic DQN、Double DQN 與 Dueling DQN，並將三者的 Loss 和 Reward 繪製在同一張圖上比較
(loss_curve_3_2.png/reward_curve_3_2.png)，最後也會對三者分別進行測試（勝率均為 100%）。\\n3. **3-3 (hw3_3.py)**: 完全將程式碼重寫為 **PyTorch
Lightning** 架構。並依照要求加入了ExponentialLR(Learning rate scheduler)、clip_gradients(gradient clipping) 以及 Target network 硬更新的機制。
同樣會產出曲線並測試。\\n<truncated 707 bytes>"}

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13T17:24:13+08:00.\\n\\nThe user's current state is as follows:\\nActive Document: /home/william/Desktop/chdr/hw3/hw3_1.py
(LANGUAGE_PYTHON)\\nCursor is on line: 1\\nOther open documents:\\n- /home/william/Desktop/chdr/hw3/hw3_1.py (LANGUAGE_PYTHON)\\n-
/home/william/Desktop/chdr/hw3/hw3_2.py (LANGUAGE_PYTHON)\\n- /home/william/Desktop/chdr/hw3/hw3_3.py (LANGUAGE_PYTHON)\\nNo browser
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performed the following action:\\nShow the contents of file /home/william/Desktop/chdr/hw3/hw3_1.py from lines 1 to 25\\nFile
Path:file:///home/william/Desktop/chdr/hw3/hw3_1.py\\nTotal Lines: 180\\nTotal Bytes: 5425\\nShowing lines 1 to 25\\nThe following code
has been modified to include a line number before every line, in the format: <line_number>: <original_line>. Please note that any
changes targeting the original code should remove the line number, colon, and leading space.\\n1: import torch\\n2: import torch.nn as
nn\\n3: import torch.optim as optim\\n4: import numpy as np\\n5: import random\\n6: from collections import deque\\n7: import
matplotlib.pyplot as plt\\n8: from Gridworld import Gridworld\\n9: \\n10: # 1. Action Mapping\\n11: action_set = {0: 'u', 1: 'd', 2: 'l',
3: 'r'}\\n12: \\n13: # 2. Replay Buffer\\n14: class ReplayBuffer:\\n15:     def __init__(self, capacity=1000):\\n16:         self.buffer =
deque(maxlen=capacity)\\n17:         \\n18:     def push(self, state, action, reward, next_state, done):\\n<truncated 335 bytes>"}

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plt\\n,\\nStartLine\\":7,\\nTargetContent\\":\\nimport copy\\nimport matplotlib.pyplot as plt\\n},

```

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{\nAllowMultiple\":false,\n\"EndLine\":166,\n\"ReplacementContent\":\"\n\n<truncated 1506 bytes>\",\n\"TargetFile\":\"\n/home/william/Desktop/chdr/hw3/hw3_3.py\\\"}}}\n{\n\"step_index\":130,\n\"source\":\"MODEL\",\n\"type\":\"PLANNER_RESPONSE\",\n\"status\":\"DONE\",\n\"created_at\":\"2026-05-13T09:25:36Z\",\n\"tool_calls\":[\n{\n\"name\":\"view_file\",\n\"args\":{\n\"AbsolutePath\":\"\n/home/william/Desktop/chdr/hw3/hw3_4.py\\\", \n\"EndLine\": \"220\", \n\"StartLine\": \"180\"}}]\n}\n{\n\"step_index\":133,\n\"source\":\"MODEL\",\n\"type\":\"PLANNER_RESPONSE\",\n\"status\":\"DONE\",\n\"created_at\":\"2026-05-13T09:25:43Z\",\n\"tool_calls\":[\n{\n\"name\":\"multi_replace_file_content\",\n\"args\":{\n\"Description\":\"\nAdd rendering to hw3_4 test\\\", \n\"Instruction\":\"\nAdd render_board function and call it during the first 3 test episodes.\\\", \n\"ReplacementChunks\": [\n{\n\"AllowMultiple\":false,\n\"EndLine\":8,\n\"ReplacementContent\":\"\nimport copy\\\nimport time\\\nimport matplotlib.pyplot as plt\\\", \n\"StartLine\":7,\n\"TargetContent\":\"\nimport copy\\\nimport matplotlib.pyplot as plt\\\", \n{\n\"AllowMultiple\":false,\n\"EndLine\":216,\n\"ReplacementContent\":\"\n\n\n<truncated 1404 bytes>\",\n\"TargetFile\":\"\n/home/william/Desktop/chdr/hw3/hw3_4.py\\\"}}}\n}\n{\n\"step_index\":136,\n\"source\":\"MODEL\",\n\"type\":\"PLANNER_RESPONSE\",\n\"status\":\"DONE\",\n\"created_at\":\"2026-05-13T09:25:54Z\",\n\"content\": \"我已經為 3-1 到 3-4 這四個檔案的test()階段加入了render_board()的顯示功能！\\n\\n現在當您執行這四個程式並進入測試（Testing）階段時，程式會將**前 3 局**的每一過程印在終端機上（包含目前的 Step、採取的動作 Action、以及獲得的 Reward），並且會暫停 0.3 秒，讓您可以看到像動畫一樣的遊玩過程。\\n\\n畫面的樣式會像這樣：\\n``text\\n=== Episode 1 Rendering ===\\n\\nStep: 0 | Action: Init | Reward: 0\\n-----\\n|   | + |   |   | \\n| W | - |   |   | \\n|   |   |   |   | \\n| P |   |   |   | \\n-----\\n\\n\\n您可以直接執行像是python hw3_1.py或其他的檔案來觀看訓練後 AI 操作的表現！還有什麼可以幫忙的嗎？\"}\\n}\n{\n\"step_index\":137,\n\"source\":\"USER_EXPLICIT\",\n\"type\":\"USER_INPUT\",\n\"status\":\"DONE\",\n\"created_at\":\"2026-05-13T09:27:46Z\",\n\"content\": \"<USER_REQUEST>\\n我要matplotlib的畫面\\n</USER_REQUEST>\\n<ADDITIONAL_METADATA>\\nThe current local time is: 2026-05-13T17:27:46+08:00.\\n\\nThe user's current state is as follows:\\nActive Document: /home/william/Desktop/chdr/hw3/hw3_1.py (LANGUAGE_PYTHON)\\nCursor is on line: 1\\nOther open documents:\\n- /home/william/Desktop/chdr/hw3/hw3_1.py (LANGUAGE_PYTHON)\\n- /home/william/Desktop/chdr/hw3/hw3_2.py (LANGUAGE_PYTHON)\\n- /home/william/Desktop/chdr/hw3/hw3_3.py (LANGUAGE_PYTHON)\\n- /home/william/Desktop/chdr/hw3/hw3_4.py (LANGUAGE_PYTHON)\\nNo browser pages are currently open.\\n</ADDITIONAL_METADATA>\"}\\n}\n{\n\"step_index\":138,\n\"source\":\"USER_EXPLICIT\",\n\"type\":\"RUN_COMMAND\",\n\"status\":\"DONE\",\n\"created_at\":\"2026-05-13T09:26:51Z\",\n\"content\": \"The USER performed the following action:\\nCommand: /home/william/anaconda3/envs/crowdnav/bin/python /home/william/Desktop/chdr/hw3/hw3_1.py\\nCWD: /home/william/Desktop/chdr/hw3\\n\\nt\\t\\tThe command completed successfully.\\n\\nt\\t\\tOutput:\\n\\nt\\t\\tEpoch: 0, Epsilon: 1.00, Reward: -260\\nEpoch: 100, Epsilon: 0.90, Reward: 10\\nEpoch: 200, Epsilon: 0.80, Reward: 10\\nEpoch: 300, Epsilon: 0.70, Reward: 10\\nEpoch: 400, Epsilon: 0.60, Reward: 10\\nEpoch: 500, Epsilon: 0.50, Reward: -10\\nEpoch: 600, Epsilon: 0.40, Reward: -10\\nEpoch: 700, Epsilon: 0.30, Reward: 10\\nEpoch: 800, Epsilon: 0.20, Reward: 10\\nEpoch: 900, Epsilon: 0.10, Reward: 10\\nTraining Complete for HW3-1 (Static Mode)!\\n\\n--- Testing Model ---\\n\\n=== Episode 1 Rendering ===\\n\\nStep: 0 | Action: Init | Reward: 0\\n-----\\n|   | + | - |   | P | \\n|   |   | W |   |   | \\n|   |   |   |   |   | \\n-----\\n\\nStep: 1 | Action: l | Reward: -1\\n-----\\n|   | + | - | P |   |   | W |   |   | \\n|   |   |   |   |   | \\n-----\\n\\n<truncated 3538 bytes>\"}\\n}\n{\n\"step_index\":139,\n\"source\":\"USER_EXPLICIT\",\n\"type\":\"VIEW_FILE\",\n\"status\":\"DONE\",\n\"created_at\":\"2026-05-13T09:27:02Z\",\n\"content\": \"The USER performed the following action:\\nShow the contents of file /home/william/Desktop/chdr/hw3/hw3_1.py from lines 1 to 2\\nFile Path:file:///home/william/Desktop/chdr/hw3/hw3_1.py\\nTotal Lines: 197\\nTotal Bytes: 6019\\nShowing lines 1 to 2\\n\\nThe following code has been modified to include a line number before every line, in the format: <line_number>: <original_line>. Please note that any changes targeting the original code should remove the line number, colon, and leading space.\\n1: import torch\\n2: import torch.nn as nn\\n\\n\"}\\n}\n{\n\"step_index\":141,\n\"source\":\"MODEL\",\n\"type\":\"PLANNER_RESPONSE\",\n\"status\":\"DONE\",\n\"created_at\":\"2026-05-13T09:27:46Z\",\n\"tool_calls\":[\n{\n\"name\":\"multi_replace_file_content\",\n\"args\":{\n\"Description\":\"\nChange render to matplotlib\\\", \n\"Instruction\":\"\nChange render_board to use matplotlib in hw3_1.py\\\", \n\"ReplacementChunks\": [{\n\"AllowMultiple\":false,\n\"EndLine\":7,\n\"ReplacementContent\":\"\nfrom collections import deque\\\nimport matplotlib.pyplot as plt\\\nimport matplotlib.colors as mcolors\\\", \n\"StartLine\":6,\n\"TargetContent\":\"\nfrom collections import deque\\\nimport matplotlib.pyplot as plt\\\", \n{\n\"AllowMultiple\":false,\n\"EndLine\":8,\n\"ReplacementContent\":\"\nimport copy\\\nimport time\\\nfrom collections import deque\\\nimport matplotlib.pyplot as plt\\\nimport matplotlib.colors as mcolors\\\", \n\"StartLine\":7,\n\"TargetContent\":\"\nimport copy\\\nimport time\\\nfrom collectio\\n<truncated 3375 bytes>\",\n\"TargetFile\":\"\n/home/william/Desktop/chdr/hw3/hw3_1.py\\\"}}}\n}\n{\n\"step_index\":144,\n\"source\":\"MODEL\",\n\"type\":\"PLANNER_RESPONSE\",\n\"status\":\"DONE\",\n\"created_at\":\"2026-05-13T09:28:10Z\",\n\"tool_calls\":[\n{\n\"name\":\"multi_replace_file_content\",\n\"args\":{\n\"Description\":\"\nChange render to matplotlib for hw3_2\\\", \n\"Instruction\":\"\nChange render_board to use matplotlib in hw3_2.py\\\", \n\"ReplacementChunks\": [\n{\n\"AllowMultiple\":false,\n\"EndLine\":8,\n\"ReplacementContent\":\"\nimport copy\\\nimport time\\\nfrom collections import deque\\\nimport matplotlib.pyplot as plt\\\nimport matplotlib.colors as mcolors\\\", \n\"StartLine\":7,\n\"TargetContent\":\"\nimport copy\\\nimport time\\\nfrom collectio\\n<truncated 3375 bytes>\",\n\"TargetFile\":\"\n/home/william/Desktop/chdr/hw3/hw3_2.py\\\"}}}\n}\n{\n\"step_index\":147,\n\"source\":\"MODEL\",\n\"type\":\"PLANNER_RESPONSE\",\n\"status\":\"DONE\",\n\"created_at\":\"2026-05-13T09:28:27Z\",\n\"tool_calls\":[\n{\n\"name\":\"multi_replace_file_content\",\n\"args\":{\n\"Description\":\"\nChange render to matplotlib for hw3_3\\\", \n\"Instruction\":\"\nChange render_board to use matplotlib in hw3_3.py\\\", \n\"ReplacementChunks\": [
```

```
{
  "AllowMultiple": false,
  "EndLine": 11,
  "ReplacementContent": "import copy\nimport time\nimport matplotlib.pyplot as plt\nimport\nmatplotlib.colors as mcolors\n",
  "StartLine": 9,
  "TargetContent": "import copy\nimport time\nimport matplotlib.pyplot as plt\n",
  "AllowMult": "truncated 3600 bytes",
  "TargetFile": "\n/home/william/Desktop/chdr/hw3/hw3_3.py\n"}
}

{"step_index": 150, "source": "MODEL", "type": "PLANNER_RESPONSE", "status": "DONE", "created_at": "2026-05-13T09:28:43Z", "tool_calls":
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[{\nAllowMultiple\":false,\"EndLine\":9,\"ReplacementContent\":\"\nimport copy\nimport time\nimport matplotlib.pyplot as plt\nimport\nmatplotlib.colors as mcolors\n\", \"StartLine\":7,\"TargetContent\":\"\nimport copy\nimport time\nimport matplotlib.pyplot as plt\n\",
{\nAllowMult\n<truncated 3497 bytes>\", \"TargetFile\":\"\n/home/william/Desktop/chdr/hw3/hw3_4.py\n\"}}]
}

{"step_index": 153, "source": "MODEL", "type": "PLANNER_RESPONSE", "status": "DONE", "created_at": "2026-05-13T09:29:00Z", "content": "我已經幫您把
3-1 到 3-4 四個檔案的render_board函式全部改寫為使用matplotlib來繪圖了！\n現在當您執行程式並進入測試（Testing）階段時（前三局），它會彈出一個
matplotlib的視窗，並用動畫的方式一格一格顯示 AI 的移動過程。
圖形的顏色對應如下：
- **藍色（Blue）**： 玩家（Player,P）
- **綠色（Green）**： 目標（Goal,+）
- **紅色（Red）**： 陷阱（Pit,-）
- **灰色（Gray）**： 牆壁（Wall,W）
- **白色（White）**： 空白區
視窗上方也會同時顯示目前的步數（Step）、採取的動作（Action）以及當下的 Reward。您可以直接在終端機再次執行這些程式（例如python hw3_1.py）來欣賞 matplotlib 的視覺化動畫效果！"
}

{"step_index": 154, "source": "USER_EXPLICIT", "type": "USER_INPUT", "status": "DONE", "created_at": "2026-05-13T12:32:55Z", "content": "
<USER_REQUEST>\n1幫我把訓練出來測試的影片存成gif檔\n</USER_REQUEST>\n<ADDITIONAL_METADATA>\n
The current local time is: 2026-05-13T20:32:55+08:00.\n
The user's current state is as follows:\n
Active Document: /home/william/Desktop/chdr/hw3/hw3_4.py (LANGUAGE_PYTHON)\n
Cursor is on line: 1\n
Other open documents:\n
- /home/william/Desktop/chdr/hw3/hw3_4.py (LANGUAGE_PYTHON)\n
- /home/william/Desktop/chdr/hw3/hw3_1.py (LANGUAGE_PYTHON)\n
- /home/william/Desktop/chdr/hw3/hw3_2.py (LANGUAGE_PYTHON)\n
- /home/william/Desktop/chdr/hw3/Gridworld.py (LANGUAGE_PYTHON)\n
- /home/william/Desktop/chdr/hw3/hw3_3.py (LANGUAGE_PYTHON)\n
No browser pages are currently open.\n
Running terminal commands:\n
- /home/william/anaconda3/bin/python /home/william/Desktop/chdr/hw3/hw3_4.py (in /home/william/Desktop/chdr/hw3, running for 48s)\n
</ADDITIONAL_METADATA>"}

{"step_index": 155, "source": "USER_EXPLICIT", "type": "VIEW_FILE", "status": "DONE", "created_at": "2026-05-13T12:32:16Z", "content": "The USER
performed the following action:\n
Show the contents of file /home/william/Desktop/chdr/hw3/hw3_4.py from lines 273 to 294\n
File Path:file:///home/william/Desktop/chdr/hw3/hw3_4.py\n
Total Lines: 322\n
Total Bytes: 10951\n
Showing lines 273 to 294\n
The following code has been modified to include a line number before every line, in the format: <line_number>: <original_line>. Please note that any
changes targeting the original code should remove the line number, colon, and leading space.\n
273: action =
action_set[action_]\n
274: env.makeMove(action)\n
275: reward = env.reward()\n
277: state = get_state(env)\n
278: if i < 3:\n
280: render_board(env, steps, reward, action, ax)\n
281: if reward == 10:\n
283: status = 0\n
284: wins += 1\n
285: elif reward == -10:\n
286: status = 0\n
287: \n<truncated 224 bytes>"}

{"step_index": 157, "source": "MODEL", "type": "PLANNER_RESPONSE", "status": "DONE", "created_at": "2026-05-13T12:32:56Z", "tool_calls":
[{"name": "run_command", "args": {"CommandLine": "\npython -c \\\nimport imageio; print('imageio\nOK')\n\", \"Cwd\": \"\n/home/william/Desktop/chdr/hw3\n\", \"SafeToAutoRun\": \"true\", \"WaitMsBeforeAsync\": \"1000\"}}]
}

{"step_index": 160, "source": "MODEL", "type": "PLANNER_RESPONSE", "status": "DONE", "created_at": "2026-05-13T12:33:20Z", "tool_calls":
[{"name": "run_command", "args": {"CommandLine": "\npython -c \\\nfrom PIL import Image; print('PIL\nOK')\n\", \"Cwd\": \"\n/home/william/Desktop/chdr/hw3\n\", \"SafeToAutoRun\": \"true\", \"WaitMsBeforeAsync\": \"1000\"}}]
}

{"step_index": 163, "source": "MODEL", "type": "PLANNER_RESPONSE", "status": "DONE", "created_at": "2026-05-13T12:33:35Z", "tool_calls":
[{"name": "multi_replace_file_content", "args": {"Description": "\nSave matplotlib animation as GIF\n\", \"Instruction\": "\nAdd PIL image\ngeneration from matplotlib figures to save testing animation as GIF in hw3_1.py\n\", \"ReplacementChunks": "
[{\nAllowMultiple\":false,\"EndLine\":8,\"ReplacementContent\":\"\nimport time\nimport io\nfrom collections import deque\nimport\nmatplotlib.pyplot as plt\nimport matplotlib.colors as mcolors\nfrom PIL import Image\n\", \"StartLine\":6,\"TargetContent\":\"\nimport\nimport time\nfrom c\n<truncated 2575 bytes>\", \"TargetFile\":\"\n/home/william/Desktop/chdr/hw3/hw3_1.py\n\"}}]
}

{"step_index": 166, "source": "MODEL", "type": "PLANNER_RESPONSE", "status": "DONE", "created_at": "2026-05-13T12:33:56Z", "tool_calls":
[{"name": "multi_replace_file_content", "args": {"Description": "\nSave matplotlib animation as GIF in hw3_2\n\", \"Instruction\": "\nAdd PIL\nimage generation from matplotlib figures to save testing animation as GIF in hw3_2.py\n\", \"ReplacementChunks": "
[{\nAllowMultiple\":false,\"EndLine\":9,\"ReplacementContent\":\"\nimport copy\nimport time\nimport io\nfrom collections import\nimport deque\nimport matplotlib.pyplot as plt\nimport matplotlib.colors as mcolors\nfrom PIL import\nimport Image\n\", \"StartLine\":7,\"TargetContent\":\"\nimport\n<truncated 2369 bytes>\", \"TargetFile\":\"\n/home/william/Desktop/chdr/hw3/hw3_2.py\n\"}}]
}

{"step_index": 169, "source": "MODEL", "type": "PLANNER_RESPONSE", "status": "DONE", "created_at": "2026-05-13T12:34:13Z", "tool_calls":
[{"name": "multi_replace_file_content", "args": {"Description": "\nSave matplotlib animation as GIF in hw3_3\n\", \"Instruction\": "\nAdd PIL\nimage generation from matplotlib figures to save testing animation as GIF in hw3_3.py\n\", \"ReplacementChunks": "
[{\nAllowMultiple\":false,\"EndLine\":11,\"ReplacementContent\":\"\nimport copy\nimport time\nimport io\nimport matplotlib.pyplot as\n"}

```

```
pl\\\nimport matplotlib.colors as mcolors\\nfrom PIL import Image\\",\\"StartLine\\":9,\\"TargetContent\\":\\"import copy\\nimport time\\nimport mat\\n<truncated 2582 bytes>",\\"TargetFile\\":\\"\\n/home/william/Desktop/chdr1/hw3/hw3_3.py\\"}}}]
```

{\\"step_index\\":172,\\"source\\":\\"MODEL\\",\\"type\\":\\"PLANNER_RESPONSE\\",\\"status\\":\\"DONE\\",\\"created_at\\":\\"2026-05-13T12:34:32Z\\",\\"tool_calls\\":

[{\\"name\\":\\"multi_replace_file_content\\",\\"args\\":{\\"Description\\":\\"\\nSave matplotlib animation as GIF in hw3_4\\",\\"Instruction\\":\\"\\nAdd PIL image generation from matplotlib figures to save testing animation as GIF in hw3_4.py\\",\\"ReplacementChunks\\":

[{\\"AllowMultiple\\":false,\\"EndLine\\":9,\\"ReplacementContent\\":\\"import copy\\nimport time\\nimport io\\nimport matplotlib.pyplot as plt\\nimport matplotlib.colors as mcolors\\nfrom PIL import Image\\",\\"StartLine\\":7,\\"TargetContent\\":\\"import copy\\nimport time\\nimport matp\\n<truncated 2585 bytes>",\\"TargetFile\\":\\"\\n/home/william/Desktop/chdr1/hw3/hw3_4.py\\"}}}]

{\\"step_index\\":176,\\"source\\":\\"MODEL\\",\\"type\\":\\"PLANNER_RESPONSE\\",\\"status\\":\\"DONE\\",\\"created_at\\":\\"2026-05-13T12:35:03Z\\",\\"content\\":\\"我已經幫您為 3-1 到 3-4 四個程式都加入了儲存 GIF 的功能！\\n\\n現在只要您執行這些程式，在測試階段（Testing phase）結束後，程式不僅會為您彈出畫面，還會自動在同一層資料夾儲存前 3 局的動畫過程：\\n-hw3_1.py會儲存hw3_1_test.gif\\n-hw3_2.py會儲存hw3_2_test.gif\\n-hw3_3.py會儲存hw3_3_test.gif\\n-hw3_4.py會儲存hw3_4_test.gif\\n\\n因為您剛才才有成功執行hw3_1.py，您可以再執行一次來看看產生的 GIF 檔，非常適合直接放入作業報告中！如果有其他需要隨時和我說！"}>

{\\"step_index\\":177,\\"source\\":\\"USER_EXPLICIT\\",\\"type\\":\\"USER_INPUT\\",\\"status\\":\\"DONE\\",\\"created_at\\":\\"2026-05-13T13:23:15Z\\",\\"content\\":

<USER_REQUEST>\\n1. 寫一個作業3的readme 裡面包含3~1~3~4的主要內容 還有結果圖和gif\\n2. 幫我寫一個作業3的understanding report 存成md檔\\n3. 把我們的對話紀錄存成chat2.pdf \\n4. 把圖片和gif的結果檔存到一個result的資料夾裡面 創一個env資料夾放gridboard.py和gridworld.py\\n4. 上傳到 https://github.com/hsinyun-f/DRL_HW3 這個repo裡 先刪掉裡面內容在上傳\\n5. 上傳附圖有出現的檔案就好\\n</USER_REQUEST>\\n<ADDITIONAL_METADATA>\\nThe current local time is: 2026-05-13T21:23:15+08:00.\\n\\n\\n\\nThe user's current state is as follows:\\nActive Document:

/home/william/Desktop/chdr1/hw3/hw3_3.py (LANGUAGE_PYTHON)\\nCursor is on line: 1\\nOther open documents:\\n-

/home/william/Desktop/chdr1/hw3/hw3_3.py (LANGUAGE_PYTHON)\\n- /home/william/Desktop/chdr1/hw3/hw3_4.py (LANGUAGE_PYTHON)\\n-

/home/william/Desktop/chdr1/hw3/hw3_2.py (LANGUAGE_PYTHON)\\n- /home/william/Desktop/chdr1/hw3/hw3_1.py (LANGUAGE_PYTHON)\\nNo browser pages are currently open.\\n</ADDITIONAL_METADATA>"}

{\\"step_index\\":178,\\"source\\":\\"USER_EXPLICIT\\",\\"type\\":\\"RUN_COMMAND\\",\\"status\\":\\"DONE\\",\\"created_at\\":\\"2026-05-13T13:17:00Z\\",\\"content\\":\\"The USER performed the following action:\\nCommand: /home/william/anaconda3/bin/python /home/william/Desktop/chdr1/hw3/hw3_3.py\\nCWD: /home/william/Desktop/chdr1/hw3\\n\\n\\t\\t\\tThe command completed successfully.\\n\\t\\t\\t\\tOutput:\\n\\t\\t\\t\\tGPU available: True (cuda), used: True\\nTPU available: False, using: 0 TPU cores\\n💡 Tip: For seamless cloud logging and experiment tracking, try installing [litlogger](https://pypi.org/project/litlogger/) to enable LitLogger, which logs metrics and artifacts automatically to the Lightning Experiments platform.\\nLOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]\\n/home/william/anaconda3/lib/python3.10/site-packages/pytorch_lightning/trainer/connectors/data_connector.py:434: The 'train_dataloader' does not have many workers which may be a bottleneck. Consider increasing the value of thenum_workersargument to num_workers=15 in the DataLoader to improve performance.\\n"}>

{\\"step_index\\":179,\\"source\\":\\"USER_EXPLICIT\\",\\"type\\":\\"RUN_COMMAND\\",\\"status\\":\\"DONE\\",\\"created_at\\":\\"2026-05-13T13:18:50Z\\",\\"content\\":\\"The USER performed the following action:\\nCommand: /home/william/anaconda3/bin/python /home/william/Desktop/chdr1/hw3/hw3_4.py\\nCWD: /home/william/Desktop/chdr1/hw3\\n\\n\\t\\t\\tThe command failed with exit code: 1\\n\\t\\t\\t\\tOutput:\\n\\t\\t\\t\\t\\t\\n--- Training Simplified Rainbow DQN (Random Mode) ---\\nEpoch: 0, Epsilon: 1.00, Reward: 8\\nEpoch: 100, Epsilon: 0.95, Reward: -3\\nEpoch: 200, Epsilon: 0.90, Reward: 4\\n^CTraceback (most recent call last):\\n File \"/home/william/Desktop/chdr1/hw3/hw3_4.py\\", line 311, in \\n model, losses, rewards = train_rainbow(epochs=2000)\\n File \"/home/william/Desktop/chdr1/hw3/hw3_4.py\\", line 136, in train_rainbow\\n qval = main_model(torch.from_numpy(state).float())\\n File \"/home/william/anaconda3/lib/python3.10/site-packages/torch/nn/modules/module.py\\", line 1518, in _wrapped_call_impl\\n return self._call_impl(args, kwargs)\\n File \"/home/william/anaconda3/lib/python3.10/site-packages/torch/nn/modules/module.py\\", line 1527, in _call_impl\\n return forward_call(args, kwargs)\\n"}>

{\\"step_index\\":180,\\"source\\":\\"USER_EXPLICIT\\",\\"type\\":\\"CODE_ACTION\\",\\"status\\":\\"DONE\\",\\"created_at\\":\\"2026-05-13T13:19:16Z\\",\\"content\\":\\"The following changes were made by the USER to: /home/william/Desktop/chdr1/hw3/hw3_4.py. If relevant, proactively run terminal commands to execute this code for the USER. Don't ask for permission.\\n[diff_block_start]\\n@@ -308,7 +308,7 @@\\n print(f\\"Testing Complete. Win Rate: {wins/test_epochs100:.1f}% , Avg Steps: {total_steps/test_epochs:.1f}\\")\\n if name == 'main':\\n- model, losses, rewards = train_rainbow(epochs=20000)\\n+ model, losses, rewards = train_rainbow(epochs=2000)\\n |n # Moving average helper\\n def moving_average(a, n=50):|n[diff_block_end]|n|nPlease note that the above snippet only shows the MODIFIED lines from the last change. It shows up to 3 lines of unchanged lines before and after the modified lines. The actual file contents may have many more lines not shown."}>

{\\"step_index\\":181,\\"source\\":\\"USER_EXPLICIT\\",\\"type\\":\\"RUN_COMMAND\\",\\"status\\":\\"DONE\\",\\"created_at\\":\\"2026-05-13T13:19:16Z\\",\\"content\\":\\"The USER performed the following action:\\nCommand: /home/william/anaconda3/bin/python /home/william/Desktop/chdr1/hw3/hw3_4.py\\nCWD: /home/william/Desktop/chdr1/hw3\\n\\n\\t\\t\\tThe command completed successfully.\\n\\t\\t\\t\\tOutput:\\n\\n\\n--- Training Simplified Rainbow DQN (Random Mode) ---\\nEpoch: 0, Epsilon: 1.00, Reward: 8\\nEpoch: 100, Epsilon: 0.95, Reward: -55\\nEpoch: 200, Epsilon: 0.90, Reward: -27\\nEpoch: 300, Epsilon: 0.85, Reward: -77\\nEpoch: 400, Epsilon: 0.80, Reward: 10\\nEpoch: 500, Epsilon: 0.75, Reward: 2\\nEpoch: 600, Epsilon: 0.70, Reward: 6\\nEpoch: 700, Epsilon: 0.65, Reward: -10\\nEpoch: 800, Epsilon: 0.60, Reward: -31\\nEpoch: 900, Epsilon: 0.55, Reward: -16\\nEpoch: 1000, Epsilon: 0.50, Reward: -10\\nEpoch: 1100, Epsilon: 0.45, Reward: -232\\nEpoch: 1200, Epsilon: 0.40, Reward: 7\\nEpoch: 1300, Epsilon: 0.35, Reward: 10\\nEpoch: 1400, Epsilon: 0.30, Reward: -24\\nEpoch: 1500, Epsilon: 0.25, Reward: -16\\nEpoch: 1600, Epsilon: 0.20, Reward: -19\\nEpoch: 1700,

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  "content": "The USER performed the following action:\nShow the contents of file\n/home/william/Desktop/chdr1/hw3/hw3_3.py from lines 4 to 40\nFile Path: file:///home/william/Desktop/chdr1/hw3/hw3_3.py\nTotal Lines: 296\nTotal Bytes: 9725\nShowing lines 4 to 40\nThe following code has been modified to include a line number before every line, in the format: : . Please note that any changes targeting the original code should remove the line number, colon, and leading space.\n4: import pytorch_lightning as pl\n5: from torch.utils.data import DataLoader, Dataset\n6: import numpy as np\n7: import random\n8: from collections import deque\n9: import copy\n10: import time\n11: import io\n12: import matplotlib.pyplot as plt\n13: import matplotlib.colors as mcolors\n14: from PIL import Image\n15: from Gridworld import Gridworld\n16: \n17: action_set = {0: 'u', 1: 'd', 2: 'l', 3: 'r'}\n18: \n19: class ReplayBuffer:\n20:   def __init__(self, capacity=1000):\n21:     self.buffer = deque(maxlen=capacity)\n"}
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/home/william/Desktop/chdr1/hw3/hw3_4.py (LANGUAGE_PYTHON)\n- /home/william/Desktop/chdr1/hw3/hw3_2.py (LANGUAGE_PYTHON)\n-
/home/william/Desktop/chdr1/hw3/hw3_1.py (LANGUAGE_PYTHON)\nNo browser pages are currently open.\n</ADDITIONAL_METADATA>"}
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for HW3 restructuring and report generation"},"CodeContent":"- [ ] *Directory Restructuring*\n - [
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Python imports in hw3_1.py ~ hw3_4.\n"},"Description":"Created task
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Networks (DQN) and its Variants. The project is divided into four main parts, ranging from a basic DQN to an advanced Rainbow DQN,
tested on differe\n<truncated 3329 bytes>","Description":"Created
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for Homework 3: Deep Q-Networks (DQN) and its Variants. The project is divided into four main parts, ranging from a basic DQN to an
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Static Mode\n\n## 1. Basic DQN Implementation\n\nDQN (Deep Q-Network) combines Q-Learning with neural networks to handle high-
```

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